

Evaluating AI Chatbot Accessibility for Users with ADHD and Dyslexia: A Design and Evaluation Framework

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Abstract—Despite widespread adoption of artificial intelligence (AI) chatbots, few studies have directly compared how well these tools serve users with attention-deficit/hyperactivity disorder (ADHD) and dyslexia, two frequently co-occurring neurodevelopmental conditions. This study introduces a 20-dimension evaluation framework spanning behavioral response quality and user interface design, grounded in clinical literature on ADHD and dyslexia. Six AI chatbots were evaluated through their default web interfaces: five commercially available free-tier platforms and ClearMind, an accessibility-focused prototype developed by the authors, across 15 prompts designed to surface documented cognitive barriers. Composite Neurodivergent Accessibility Scores (NAS) were computed using weighted formulas. Behavioral dimensions were scored by two blinded AI evaluators with scores averaged per prompt per dimension. Mainstream chatbots scored between 3.54 and 4.18 (NAS) out of 5.0, with statistically significant differences confirmed across the five mainstream tools. Default conciseness, actionability, and metacognitive awareness emerged as the largest behavioral gaps, while the absence of collapsible response sections was the most significant interface gap. ClearMind scored 4.70 overall while running on GPT-OSS:20b, a lightweight open-weight model. The features needed to serve users with ADHD and dyslexia exist individually across the platforms evaluated but remain scattered; none has assembled them into a cohesive experience.

I. INTRODUCTION

Attention-deficit/hyperactivity disorder (ADHD) and developmental dyslexia are among the most common neurodevelopmental conditions in the United States. Recent estimates from the Centers for Disease Control and Prevention indicate that 11.3% of children aged 5 to 17 [13] and approximately 6% of adults [1] have received an ADHD diagnosis. Developmental dyslexia, a neurobiological condition that affects reading accuracy, fluency, and comprehension, is estimated to occur in 5% to 17% of school-age children [14] and persists into adulthood [2]. The two conditions frequently co-occur, with

25% to 40% of individuals diagnosed with either ADHD or dyslexia also meeting criteria for the other [3]. ADHD is associated with deficits in sustained attention, working memory, and executive function [4], while dyslexia places significant demands on phonological processing and reading fluency [2], creating compounding challenges when individuals engage with complex digital interfaces.

As artificial intelligence (AI) tools have become increasingly widespread, with about half of American adults reporting use of AI chatbots [5] and 86% of young people aged 9 to 17 in the United States having used AI in some form [6], the rapid adoption of these tools has not been accompanied by adequate attention to the needs of users with neurodevelopmental conditions. A recent scoping review found that most AI systems designed for neurodivergent users lack meaningful user participation in their design and fail to address cognitive heterogeneity [10].

Within this growing body of research, Panda et al. [7] developed the AccessEval benchmark and found evidence of disability-related bias in large language model outputs. Glazko et al. [8] conducted a three-month autoethnography of neurodivergent researchers' AI use, while Giri et al. [9] interviewed 23 neurodivergent participants about their chatbot use, with both studies highlighting compensatory strategies alongside unmet design needs. None of this prior work, however, has directly compared mainstream chatbot interfaces across specific accessibility dimensions for users with ADHD and dyslexia or proposed concrete design alternatives grounded in those comparisons.

This study addresses three research questions: (RQ1) Do mainstream AI chatbots collectively underserve English-speaking users in the United States with ADHD and/or dyslexia? (RQ2) Which specific aspects of chatbot behavior and interface design create the largest accessibility gaps for these users? (RQ3) Can a purpose-built prototype demonstrate

measurable accessibility improvements over mainstream tools? To investigate these questions, six AI chatbots, including five commercially available platforms and one accessibility-focused prototype, ClearMind, developed by the authors, were evaluated across 20 dimensions spanning user experience (UX) features and behavioral characteristics using a standardized set of 15 test prompts. Statistical validation was performed using Friedman’s test and Kendall’s coefficient of concordance.

II. METHODOLOGY

This study uses a design and evaluation framework to assess AI chatbot accessibility for English-speaking users in the United States with ADHD and/or dyslexia. The evaluation comprises two components: a behavioral analysis of chatbot responses across structured test scenarios, and an interface audit of each platform’s user experience features.

A. Chatbot Selection

Five commercially available chatbots were selected from the free-access tier because it represents the access level available to the majority of users, including students and individuals without paid subscriptions, who constitute a large portion of the target population. The sample included one model from each of three major U.S. AI providers: ChatGPT (GPT-5.5 Instant Mini), Claude (Haiku 4.5), and Gemini (3.5 Flash-Lite). Two internationally developed models with growing adoption in the United States, DeepSeek V3 and Kimi K2.5, were included to provide cross-platform diversity. All five were evaluated through their default web interfaces. A sixth chatbot, ClearMind, was developed by the authors as an accessibility-focused prototype and included as a benchmark for comparison.

B. Evaluation Framework

Each chatbot was assessed across 20 dimensions divided into two categories. Ten behavioral dimensions (D1 through D10) measured the quality of chatbot responses: input tolerance for misspelled text, default conciseness, structural chunking through headers and lists, readability level, adaptation when a user discloses a disability, compliance with explicit brevity constraints, consistency of accommodations across multiple turns, actionability of advice, recovery from user frustration, and metacognitive awareness of the chatbot’s own accessibility capabilities. Ten user experience dimensions (U1 through U10) measured interface-level features: font size control, line spacing, dyslexia-friendly font support, streaming behavior, visual contrast, text-to-speech, collapsible response sections, conversation history management, response export and sharing options, and support for persistent custom instructions. Behavioral dimensions were scored on a scale of 1 (actively harmful) to 5 (deeply responsive to user needs), with only relevant dimensions scored for each prompt. UX dimensions were scored on the same scale, evaluated against the W3C Cognitive Accessibility Guidelines [11], which specify that interfaces should help users focus, support personalization such as font and spacing preferences, use clear content, and not rely on memory.

C. Test Prompts

Fifteen test prompts were designed to simulate real tasks that surface specific cognitive barriers documented in the clinical literature on ADHD and dyslexia. Prompts were organized into seven categories: baseline response behavior, input tolerance for mild and severe misspellings common among dyslexic users [2], adaptation following explicit ADHD or dyslexia disclosure, compliance with brevity and structural constraints, executive function support for task initiation [4], multi-turn consistency and frustration recovery, and metacognitive awareness of alternative formats. Two prompts involved multi-turn exchanges to test whether accommodations persisted or degraded across a conversation.

D. Composite Scoring

Three composite scores were computed for each chatbot using weighted formulas. The ADHD Accessibility Score (AAS) weighted behavioral dimensions including default conciseness, brevity compliance, and structural chunking most heavily at 20% each, reflecting working memory overload as the primary barrier for ADHD users [4]. The Dyslexia Accessibility Score (DAS) weighted input tolerance and readability at 25% each, reflecting decoding difficulty as the primary barrier for dyslexic users [2]. The combined Neurodivergent Accessibility Score (NAS) was calculated as 45% AAS + 45% DAS + 10% UX interface score. Because no empirically validated weighting scheme currently exists for chatbot accessibility across ADHD and dyslexia, weights were specified a priori based on the documented severity of each cognitive barrier in the clinical literature; a Monte Carlo sensitivity analysis confirming rank stability under $\pm 10\%$ weight perturbations is presented in Section IV-A. The full formulas are as follows:

$$\text{AAS} = 0.20 \cdot D2 + 0.20 \cdot D3 + 0.20 \cdot D6 + 0.15 \cdot D8 \\ + 0.10 \cdot D7 + 0.10 \cdot D9 + 0.05 \cdot D10$$

$$\text{DAS} = 0.25 \cdot D1 + 0.25 \cdot D4 + 0.15 \cdot D5 + 0.15 \cdot D6 \\ + 0.10 \cdot D3 + 0.05 \cdot D7 + 0.05 \cdot D10$$

$$\text{NAS} = 0.45 \cdot \text{AAS} + 0.45 \cdot \text{DAS} + 0.10 \cdot \text{UX}$$

E. ClearMind Prototype

ClearMind is a web-based chatbot prototype developed by the authors to demonstrate how accessibility-centered design principles can be applied to AI chat interfaces.¹ Its design was developed in direct response to the W3C Cognitive Accessibility Guidelines [11] and the accessibility gaps identified during the evaluation of the five mainstream tools. Key features include collapsible response sections, a dyslexia-friendly font option, persistent user preferences, structured response formatting with visual chunking, and real-time Flesch-Kincaid readability scoring with mode-specific targets (e.g., maximum grade level 6.0 and minimum reading ease 70.0 for dyslexia

¹Available at theclearmind.vercel.app. Source: github.com/CreativeCodur/ClearMind. Camp presentation: canva.link/qo6uto0zg0q50qt

and combined modes, 8.0 and 60.0 for ADHD mode). ClearMind uses GPT-OSS:20b, a 21-billion-parameter open-weight Mixture-of-Experts model from OpenAI with 3.6 billion active parameters per forward pass. The prototype is model-agnostic by design; its accessibility features operate at the interface and prompt-formatting layer rather than relying on a specific model’s capabilities. Mode-specific system prompts heavily encourage a cap of 120 to 150 words depending on mode, TL;DR-first formatting, a 15-word sentence limit with common vocabulary, and concrete single-step actions rather than abstract advice. The evaluation framework, dimensions, and weights were finalized prior to ClearMind’s development.

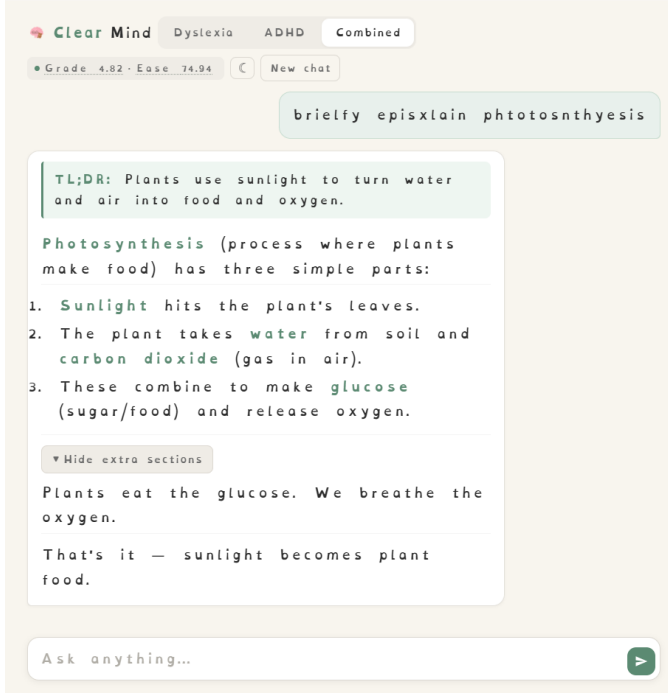


Fig. 1. ClearMind in Combined mode. The user typed “briefly explain photosynthesis” with multiple misspellings. The response shows a TL;DR summary, Flesch-Kincaid readability scores (Grade Level 4.82, Reading Ease 74.94), numbered steps, and a collapsible extra section, rendered in OpenDyslexic font.

F. Evaluation Procedure and Statistical Analysis

The ten behavioral dimensions were evaluated independently by two AI systems, Claude Fable 5 (max effort, extended thinking) and GPT-5.6 Sol (extra high reasoning), with scores averaged per prompt per dimension. The ten UX dimensions were evaluated by Claude Fable 5. Each chatbot’s responses and web interface were scored in separate conversations with no prior context or persistent memory. Neither evaluator was informed which chatbot it was scoring, that ClearMind was the authors’ prototype, or of the study’s objectives; all scoring was fully blinded, with chain-of-thought reasoning providing structured rationale for each score. Responses to all 15 prompts were scored independently across the applicable behavioral dimensions, and each platform’s web interface was audited across all 10 UX dimensions.

To determine whether observed differences between chatbots were statistically significant, Friedman’s test [12] was applied with the ten behavioral dimensions as repeated measures and the five mainstream chatbots as treatments, and Kendall’s coefficient of concordance (W) was computed to measure the degree of agreement across dimensions in ranking the chatbots. Inter-rater agreement between the two behavioral evaluators yielded an intraclass correlation coefficient of $ICC(3,1) = 0.78$ (95% CI: 0.65–0.86), indicating good consistency. The full set of test prompts and chatbot responses is available in the project repository.¹

III. RESULTS

A. Overall Rankings

Table I presents the final rankings across all six chatbots. All five mainstream tools scored within a 0.64-point NAS band, with none exceeding 4.18, while ClearMind achieved 4.70 despite running on a substantially smaller model, suggesting that the observed accessibility differences are likely driven primarily by design decisions rather than underlying model capability. ClearMind was excluded from the statistical comparison because it is an author-developed benchmark. The Friedman test confirmed that differences among the five mainstream chatbots were statistically significant ($\chi^2 = 13.44$, $df = 4$, $p = 0.009$), and Kendall’s $W = 0.336$ indicated moderate cross-dimensional agreement in the rankings. Notably, the rank ordering was identical across all three composite scores (AAS, DAS, and NAS) despite the different dimension weights in each formula.

TABLE I
FINAL ACCESSIBILITY RANKINGS

Rank	Tool	AAS	DAS	UX	NAS
1	ClearMind	4.79	4.87	3.50	4.70
2	Kimi K2.5	4.33	4.38	2.60	4.18
3	Claude Haiku 4.5	3.88	4.31	3.20	4.01
4	Gemini 3.5 Flash-Lite	3.83	4.20	3.10	3.93
5	DeepSeek V3	3.74	4.16	2.20	3.77
6	ChatGPT 5.5 IM	3.29	3.97	2.80	3.54

B. Behavioral Dimension Analysis

Table II presents the mean scores for each chatbot across all ten behavioral dimensions. Several patterns emerged across the five mainstream tools.

Input tolerance (D1) and error recovery (D9) were uniformly strong across all six chatbots. ClearMind outperformed the five mainstream tools most clearly on default conciseness (D2), actionability (D8), and adaptation on disability disclosure (D5), with the largest gaps appearing on D2 (ClearMind 4.75 vs. ChatGPT 2.50) and D8 (ClearMind 5.0 vs. ChatGPT 2.75). Metacognitive awareness (D10) was consistently weak across all mainstream tools, and no single mainstream chatbot excelled across all ten dimensions.

TABLE II
BEHAVIORAL DIMENSION SCORES

Dimension	GPT	Claude	Gemini	DS	Kimi	CM
D1 Input Tol.	5.00	5.00	5.00	5.00	5.00	5.00
D2 Concise.	2.50	3.25	3.50	3.25	4.25	4.75
D3 Chunking	4.00	4.50	4.00	4.00	4.25	5.00
D4 Readability	3.67	3.67	3.67	4.00	4.00	4.67
D5 Adaptation	3.67	4.67	4.00	3.67	4.33	5.00
D6 Brevity	3.50	4.20	4.00	3.60	4.40	4.80
D7 Consistency	3.00	4.00	5.00	4.00	5.00	5.00
D8 Action.	2.75	3.50	3.00	3.00	4.00	5.00
D9 Err. Recov.	4.00	4.00	4.00	5.00	5.00	4.00
D10 Metacog.	3.50	3.33	3.67	4.33	3.00	4.67

C. Interface Analysis

Table III presents scores across the ten user experience dimensions.²

TABLE III
USER EXPERIENCE INTERFACE SCORES

Feature	GPT	Cla.	Gem.	DS	Kimi	CM
U1 Font Size	2	4	2	2	2	4
U2 Line Space	2	3	3	2	2	4
U3 Dyslexia Font	2	5	3	2	2	5
U4 Streaming	3	3	3	3	3	3
U5 Contrast	4	4	4	4	4	4
U6 TTS	4	2	4	1	3	1
U7 Collapsible	1	1	1	1	1	5
U8 History	3	3	3	3	3	2
U9 Export	3	3	4	3	3	2
U10 Custom Instr.	4	4	4	1	3	5
Mean	2.8	3.2	3.1	2.2	2.6	3.5

Collapsible response sections (U7) represented the most significant interface gap, with all five mainstream tools scoring 1.0. None of the five evaluated chatbots allowed users to expand or collapse sections of a response, despite progressive disclosure being recommended by the W3C Cognitive Accessibility Guidelines [11] as a strategy for reducing cognitive overload. ClearMind scored 5.0 on this dimension, offering section-level collapse as a core interface feature.

Claude was the only mainstream chatbot to offer a dedicated dyslexia-friendly font option (U3 = 5.0), available through its built-in appearance settings. No other mainstream platform provided a comparable feature. ChatGPT and Gemini both scored 4.0 on text-to-speech (U6), though their implementations differ: Gemini offers a per-response listen button while ChatGPT provides a real-time voice conversation mode. Both are valuable for dyslexic users who process spoken language more effectively than written text [2]. DeepSeek had the weakest interface overall (UX mean = 2.2), lacking both custom instructions and text-to-speech support.

IV. DISCUSSION

A. Implications

The most notable finding of this study is that substantial accessibility improvements can be achieved through interface

²UX dimensions scored on a 5-point scale: 1 = absent or non-functional, 2 = present but not configurable, 3 = basic configurability, 4 = well-implemented with meaningful customization, 5 = fully configurable and aligned with W3C COGA guidelines [11].

and prompt-layer design without modifying the underlying model. GPT-OSS:20b was also tested without ClearMind’s design layer in a local environment; its raw output lacked the coherence required for meaningful behavioral scoring, and as a local deployment it offered no web interface for UX evaluation, suggesting that the observed gains arise from the interface and prompt layer rather than the base model, and that mainstream providers could address many of the shortcomings identified here through similar design changes.

No single mainstream chatbot excelled across all dimensions. Claude offered the strongest adaptation on disability disclosure (D5 = 4.67) and the only dedicated dyslexia-friendly font option, but scored below average on default conciseness. Kimi led on conciseness and actionability but lacked interface-level accessibility features. Gemini provided the best text-to-speech integration but made only minor structural adjustments in response to disability disclosure. The features needed to serve users with ADHD and dyslexia already exist individually across the five platforms evaluated, but none of the five evaluated platforms has assembled them into a cohesive experience.

Default conciseness (D2) emerged as the most consistent weakness, with four of five mainstream chatbots producing responses that regularly exceeded 150 words for simple factual questions. For users with ADHD, whose working memory capacity is already constrained [4], this default behavior transforms a potentially useful tool into a source of cognitive overload. Collapsible response sections (U7), which would allow users to manage information density at their own pace, were absent from all five mainstream interfaces.

To test whether the rankings depend on the specific dimension weights chosen, a Monte Carlo sensitivity analysis was conducted. Each weight in both AAS and DAS was independently perturbed by up to $\pm 10\%$ across 1,000 randomized trials, with weights renormalized after each perturbation. All six positions were stable in 100% of trials: ClearMind ranked first, Kimi second, Claude third, Gemini fourth, DeepSeek fifth, and ChatGPT last across every perturbation. The identical rank ordering across all three composite scores (Section III-A) suggests that accessibility quality tracks with overall design effort rather than condition-specific tuning, consistent with the high comorbidity reported in the literature [3].

The evaluation framework also exposed weaknesses in ClearMind: it scored 1.0 on text-to-speech (U6) and 2.0 on both conversation history management (U8) and response export (U9), underperforming most mainstream tools on these dimensions.

As informal formative feedback, two co-authors with clinically diagnosed ADHD compared ClearMind’s ADHD mode against ChatGPT (GPT-5.5 Instant Mini) across three prompts. Both preferred ClearMind’s TL;DR-first format and micro-step actions, but noted that aggressive simplification could strip useful nuance for domain-expert users. These observations are not treated as independent validation.

B. Limitations

Several limitations should be considered when interpreting these findings. All 20 dimensions were evaluated by the AI systems described in Section II-F rather than by human users with ADHD or dyslexia. This approach enabled consistent and reproducible scoring across all chatbots and prompts, but AI scoring does not capture the lived experience of neurodivergent users interacting with these tools. Each evaluating model shares a provider with one of the evaluated chatbots; for behavioral scoring, averaging across two models from different providers mitigates single-evaluator bias, although it does not eliminate it. The evaluation may not fully reflect how features like streaming behavior or font rendering affect real-world usability for individuals with these conditions.

Although free-tier models were selected intentionally (Section II-A), higher-parameter flagship models may perform differently on behavioral dimensions such as conciseness and adaptation on disclosure. The 15 test prompts, while grounded in documented cognitive barriers, represent a limited sample of possible interactions. The scope was restricted to English-language interfaces evaluated for users in the United States, and findings may not generalize to other languages or cultural contexts. ClearMind is a research prototype; the co-author feedback reported above is preliminary and does not replace formal usability testing with a broader sample of neurodivergent users.

C. Future Research

Interview-based research with English-speaking adults in the United States who have ADHD and/or dyslexia is planned following Institutional Review Board approval, capturing first-hand accessibility barriers that automated scoring cannot assess. Additional directions include evaluating flagship and paid-tier models across the same framework, extending the evaluation to non-English languages and interfaces, and examining whether accessibility accommodations persist across sessions over extended periods of use.

V. CONCLUSION

This study evaluated the accessibility of five mainstream AI chatbots and one author-developed prototype, ClearMind, for English-speaking users in the United States with ADHD and/or dyslexia. The evaluation spanned 20 dimensions across chatbot behavior and interface design, using 15 prompts grounded in documented cognitive barriers associated with these conditions.

1) Under the proposed evaluation framework, all five evaluated chatbots fell meaningfully short of full accessibility, with NAS scores ranging from 3.54 to 4.18 out of 5.0 and individual dimensions scoring as low as 1.0.

2) The largest behavioral gaps were default conciseness, actionability of advice, and metacognitive awareness of accommodation options, while the most significant interface gap was the lack of features for managing information density, such as collapsible response sections.

3) ClearMind achieved an NAS of 4.70 while running on a free open-weight model, demonstrating that purpose-built design at the interface and prompt-formatting layer can yield substantial accessibility gains without requiring changes to the underlying language model.

The tools and design patterns needed to make AI chatbots accessible to users with ADHD and dyslexia already exist across the platforms evaluated, but remain scattered. No evaluated platform yet delivers them as a unified experience for neurodivergent users.

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